

DevOps Lab Documentation

Program 3: Code a Dockerized python flask or [node.js](#) application

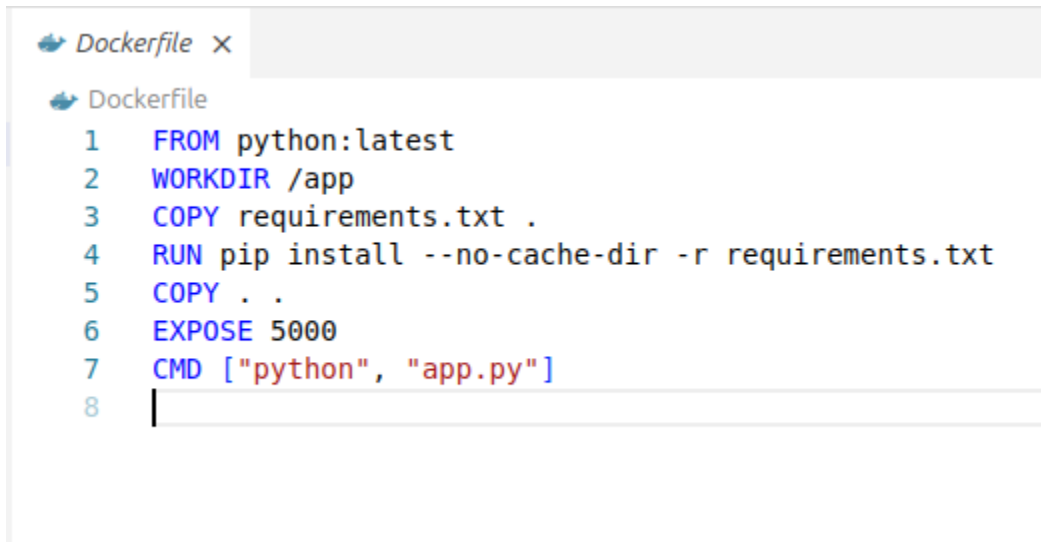
Program_3

—>Dockerfile

—>requirements.txt

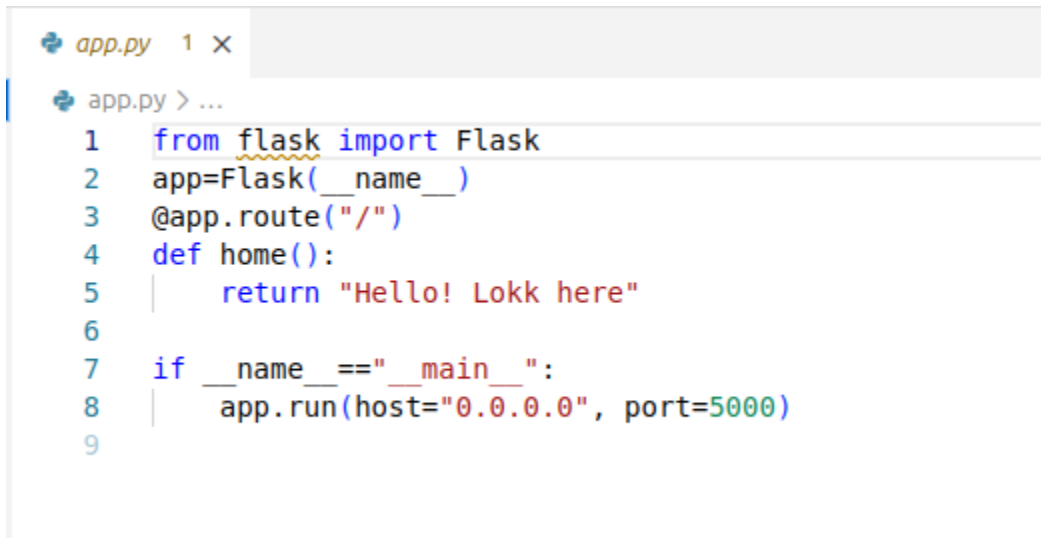
—>[app.py](#)

S1. Create a Dockerfile

A screenshot of a code editor showing a Dockerfile. The editor has a tab labeled 'Dockerfile' with a close button. The code is as follows:

```
1 FROM python:latest
2 WORKDIR /app
3 COPY requirements.txt .
4 RUN pip install --no-cache-dir -r requirements.txt
5 COPY . .
6 EXPOSE 5000
7 CMD ["python", "app.py"]
8
```

S2. Create a file [app.py](#)

A screenshot of a code editor showing an app.py file. The editor has a tab labeled 'app.py' with a close button and a file icon. The code is as follows:

```
1 from flask import Flask
2 app=Flask(__name__)
3 @app.route("/")
4 def home():
5     return "Hello! Lokk here"
6
7 if __name__=="__main__":
8     app.run(host="0.0.0.0", port=5000)
9
```

S3. Make a file requirements.txt

```
requirements.txt x
requirements.txt
1 Flask==2.3.3
2
```

S4. Build the image

docker build -t program-3 .

```
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program3$ docker build -t program-3 .
[+] Building 558.8s (13/13) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 192B
=> [internal] load metadata for docker.io/library/python:latest
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:latest@sha256:6d58c1a9444bc2664f0fa20c43a592fdb2698eb9a9c32257516538a2746c19a
=> resolve docker.io/library/python:latest@sha256:6d58c1a9444bc2664f0fa20c43a592fdb2698eb9a9c32257516538a2746c19a
=> => sha256:1edf53fea0281306069b0c05371a2ad9c5ac12e7474dc0cd6c4902c7eb33a9c9 6.49kB / 6.49kB
=> => sha256:15f14138abe4f09d3ef3953105144728056046ae469ae21e8e42749bacd67cca 25.61MB / 25.61MB
=> => sha256:6d58c1a9444bc2664f0fa20c43a592fdb2698eb9a9c32257516538a2746c19a 10.95kB / 10.95kB
=> => sha256:3b700cf7651a764f8c33f574fe4f71be54f77cf626c3b2572e26493ca465e3b1 2.32kB / 2.32kB
=> => sha256:281b80c799ded5b9a390d2e8c59930db01ee395ab809dd34259897c660751f4e 49.29MB / 49.29MB
=> => sha256:378c64c4458002be86f2d86c5768ae9ec0cff69afac7d1444e50206dc4566fa9 67.78MB / 67.78MB
=> => sha256:02e37abc533a09f86edcf3602251a61a6208ddaa131e1ald5757e5f94a54d645 235.98MB / 235.98MB
=> => sha256:b1741a62ccced4e959c504d7a6a25e533c5783994c386b24d7b31a646163283d 6.08MB / 6.08MB
=> => sha256:add107facb26b82200d59d5550cbbab6af9cf38c971f00a43e3d3cda71ffe76a 29.40MB / 29.40MB
=> => sha256:1227bf08bd420f8324f592cf370b34de4c11dd57aba693e8f3ec9ccbda8d468d 249B / 249B
=> extracting sha256:281b80c799ded5b9a390d2e8c59930db01ee395ab809dd34259897c660751f4e
=> extracting sha256:15f14138abe4f09d3ef3953105144728056046ae469ae21e8e42749bacd67cca
=> extracting sha256:378c64c4458002be86f2d86c5768ae9ec0cff69afac7d1444e50206dc4566fa9
=> extracting sha256:02e37abc533a09f86edcf3602251a61a6208ddaa131e1ald5757e5f94a54d645
=> extracting sha256:b1741a62ccced4e959c504d7a6a25e533c5783994c386b24d7b31a646163283d
=> extracting sha256:add107facb26b82200d59d5550cbbab6af9cf38c971f00a43e3d3cda71ffe76a
=> extracting sha256:1227bf08bd420f8324f592cf370b34de4c11dd57aba693e8f3ec9ccbda8d468d
=> [internal] load build context
=> => transferring context: 448B
=> [auth] library/python:pull token for registry-1.docker.io
=> [auth] library/python:pull token for registry-1.docker.io
=> [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:b4871b240a7262c1eb01ef5a80c1e9d6ff57a9f23428440fe87c8cfb3fb65e74
=> => naming to docker.io/library/program-3
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program3$
```

S6. Verify the image

docker images

```
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program3$ docker images
REPOSITORY          TAG         IMAGE ID      CREATED        SIZE
program-3           latest     b4871b240a72  8 minutes ago  1.13GB
```

S7. Run the docker run command specifying the port number

docker run -p 5000:5000 program-3

```
○ irv24mc113_varun@varun-srivastava:~/DevOpslab/program3$ docker run -p 5000:5000 program-3
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.18.0.2:5000
Press CTRL+C to quit
172.18.0.1 - - [06/Jan/2026 12:21:05] "GET / HTTP/1.1" 200 -
172.18.0.1 - - [06/Jan/2026 12:21:05] "GET /favicon.ico HTTP/1.1" 404 -
■
```

S8. Test the container by verifying the localhost details on web browser

<http://localhost:5000>

